UNIVERSITY OF WOLLONGONG AUSTRALIA

Project 2

DC Motor Speed Control

Individual project · 100 marks

MATLAB submission: plots and modified scripts for **Stage 3, Questions 2–6**.

A DC motor drives a robot joint. Model the motor, analyse its linear speed-feedback loop, then test the effect of voltage saturation. Use Lectures 6–8.

Model guide: Lecture 1, slide 26 (structure and symbols); Lecture 6, slides 12–14 (balances and transfer functions).

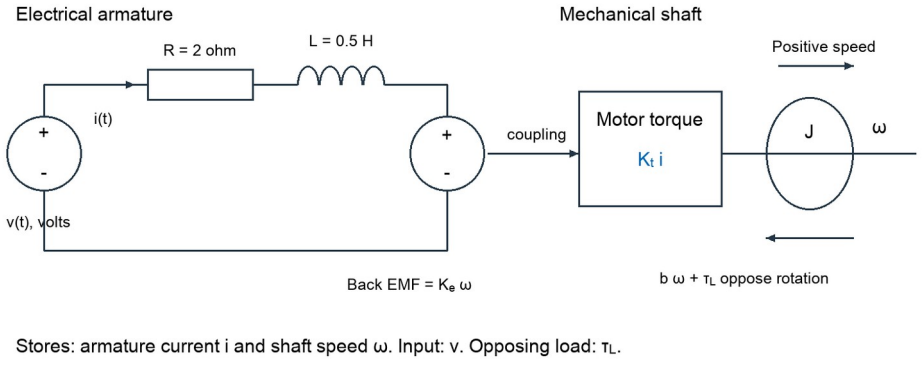


Figure 1. Supplied armature and shaft model. Current is positive into the armature; voltage polarity and positive speed are shown. Load torque opposes that speed.

Table 1. Motor parameters and test conditions.

Symbol	Value	Meaning
R	2 Ω	Armature resistance
L	0.5 H	Armature inductance
J	0.02 kg·m²	Rotor plus reflected load inertia
b	0.10 N·m·s/rad	Viscous friction
K_t	0.10 N·m/A	Torque constant
K_e	0.10 V·s/rad	Back-EMF constant
r	8 rad/s	Reference speed at t = 0
τ_L	0.40 N·m	Opposing load applied at t = 5 s

Initially, current and speed are zero. Observe the response over 0–10 s. Motor input is voltage v(t) (V); output is speed ω(t) (rad/s); current is in A.

Use $R_{ref}(s)$ for the reference transform; R denotes resistance. Current and thermal ratings are not supplied.

Stage 1 Motor model after Lecture 6 (20 marks)

The two governing laws and their coupling terms are supplied. The load is positive when it opposes positive speed.

$$L\frac{di(t)}{dt} + Ri(t) + K_e\omega(t) = v(t)$$

$$J\frac{d\omega(t)}{dt} + b\omega(t) = K_t i(t) - \tau_L(t)$$

1. Insert the numerical coefficients from Table 1.

2. Explain which term stores energy in each balance and why τ_L is subtracted.

3. With zero initial conditions and no external load torque ($\tau_L = 0$), derive the voltage-to-speed transfer function Gv(s) = Ω(s)/V(s).

4. For this question only, consider L = 0.5 H and L = 0.005 H, with all other parameters from Table 1. Calculate the electrical and effective mechanical time constants for each case (*Lecture 6, slide 14).* Explain which case permits a first-order speed approximation by neglecting inductance.

Stage 2 Fixed feedback after Lecture 7 (30 marks)

Use the **Table 1** motor in the **Figure 2** speed-feedback loop, with proportional gain K = 4 V/(rad/s).

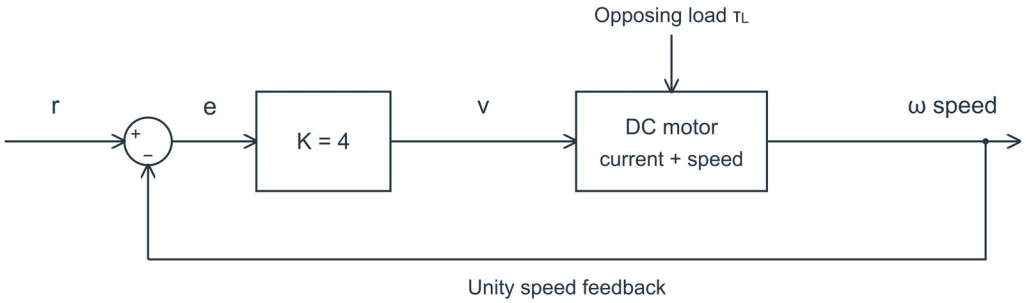


Figure 2. Linear proportional speed-control loop.

Assume zero initial conditions and keep the feedback connected. Let r(t) be the reference speed, ω(t) the motor speed and e(t) = r(t) − ω(t) the speed error.

The corresponding Laplace transforms are $R_{ref}(s)$, $\Omega(s)$, $E(s)$, respectively.

1. With zero external load torque, derive the transfer function from reference speed to motor speed: $T_r(s) = \dfrac{\Omega(s)}{R_{ref}(s)}$.

2. Derive the following two closed-loop transfer functions:

a) From reference speed to speed error: $S(s) = \dfrac{E(s)}{R_{ref}(s)}$, with zero external load torque.

b) From opposing load torque to motor speed: $T_\tau(s) = \dfrac{\Omega(s)}{T_L(s)}$, with zero reference speed.

3. Calculate the DC gains of $T_r(s)$ and $T_\tau(s)$. Using the reference speed and opposing load specified in **Table 1**, with both inputs present, use the Final Value Theorem to calculate the steady speed, error and voltage, and complete the loaded row of **Table 2**.

Table 2. No-load and loaded steady-state results.

Case	Speed rad/s	Error rad/s	Voltage V
No load (supplied)	5.25	2.75	11.0
0.40 N·m load			

4. Does the feedback loop reach the reference speed under load? Explain.

Stage 3 Read stability and test feasibility after Lecture 8 (50 marks)

3.1 Nominal linear stability and frequency response

For the linear loop from Stage 2, require phase margin PM ≥ 45° and gain margin GM ≥ 6 dB.

1. Find the closed-loop poles from your Stage 2 model. State whether the linear response is stable and whether its transient oscillates.

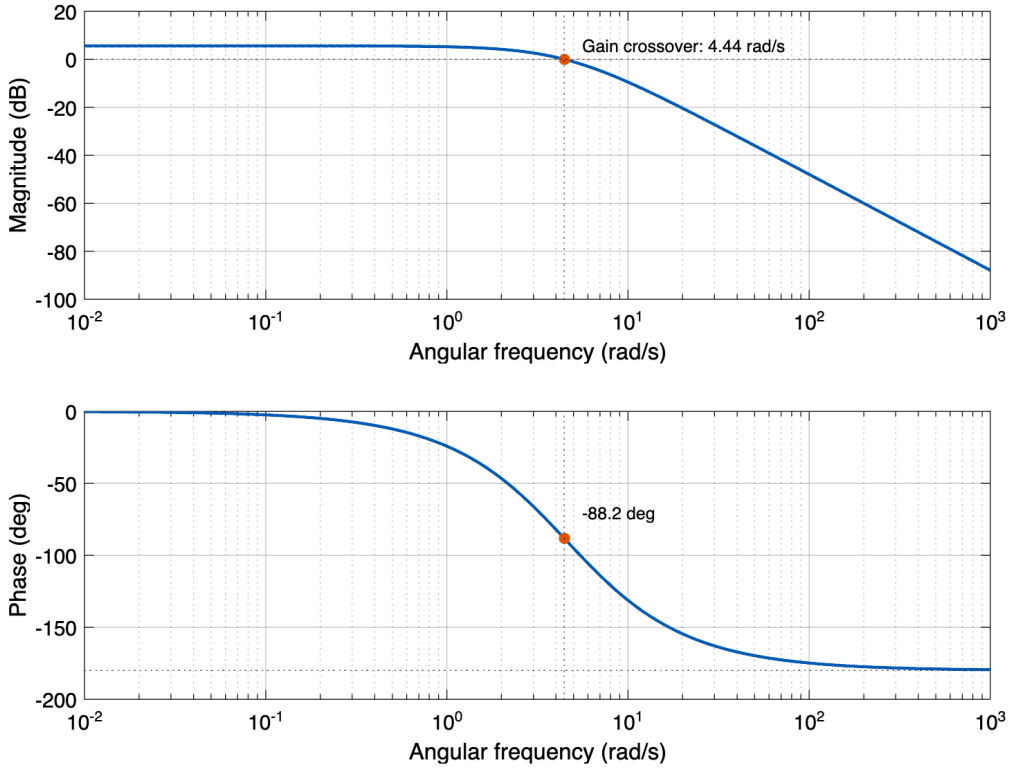


Figure 3. Nominal open-loop Bode plot.

Complete Table 3 using your model, Figure 3 and your MATLAB Nyquist plot. P and Z count open-loop and closed-loop right-half-plane poles, respectively; N is the net clockwise encirclement count of −1. Approximate Bode readings are acceptable.

Table 3. Nyquist counts and Bode readings.

Nominal record	Value
Gain crossover ω_{gc}	
Phase at gain crossover	
Finite phase crossover	
Open-loop RHP poles P	
Net clockwise encirclements N	
Closed-loop RHP poles Z	
Closed-loop stable or unstable	

2. Use MATLAB to plot the complete Nyquist curve of your nominal open-loop transfer function L(s) = K G_v(s). Include both frequency branches, mark −1 and indicate the traversal direction. Submit your plot and script. Determine P, N and Z and state whether the closed loop is stable, counting clockwise encirclements of −1 as positive and using Z = P + N. Then determine the phase and gain margins from Figure 3 and check both margin requirements.

MATLAB hint (Control System Toolbox): numL and denL are the numerator and denominator coefficient vectors of your own L(s), in descending powers of s.

```
L = tf(numL, denL);
figure; nyquist(L); grid on;
pole(L)
```

Without that toolbox, use polyval to evaluate L(jω), plot its real and imaginary parts with plot, include the complex-conjugate branch, and use roots(denL) for the open-loop poles.

3. Plot the magnitude of your Stage 2 reference-to-speed transfer function over 0.1–100 rad/s, using a logarithmic frequency axis. Report its magnitudes at 1 and 10 rad/s and submit your MATLAB script.

3.2 Voltage-limited response and target feasibility

For Questions 4–6, add the ±24 V voltage saturation in Figure 4 to the Stage 2 loop.

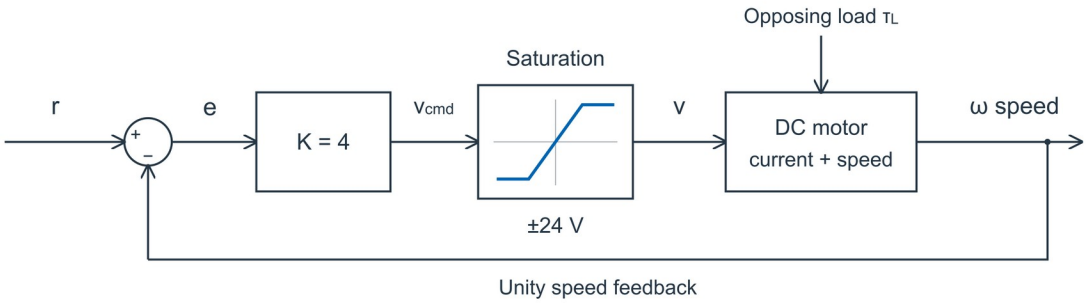


Figure 4. Speed-control loop with voltage saturation at ±24 V.

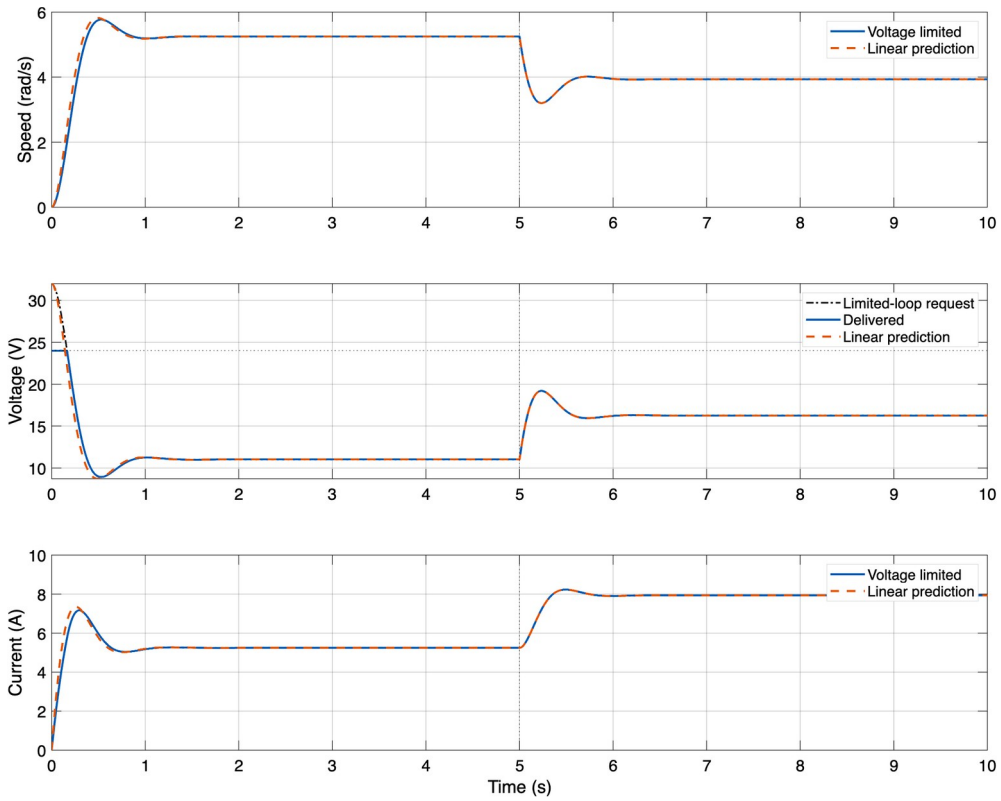


Figure 5. Supplied speed, requested/delivered voltage and current histories for K = 4. The load is applied at 5 s. Blue includes the voltage limit; the orange linear prediction has no voltage limiter. Observation window: 0–10 s.

4. Simulate the Figure 4 loop for reference speeds of 8 and 6 rad/s using Starter/Project02_starter.m. Retain the 0.40 N·m load step at 5 s. Submit one figure with speed and tracking-error panels over 0–10 s, showing both reference speeds.

5. For both reference speeds, tabulate the loaded steady speed, error and delivered voltage, and the initial requested and delivered voltages. Check the loaded results analytically.

6. Calculate the current and voltage required to hold each reference speed under load. Assess voltage feasibility and actual tracking with K = 4. Submit your modified MATLAB script.

Optional bonus Supplied state model

Optional state-space bonus: up to 10 marks, added to the required-question score; final score capped at 100. The required questions alone can earn 100 marks. Use the supplied model with state order [i, ω] and input order [v, τL].

$$\frac{dq(t)}{dt} = Aq(t) + Bu(t), \qquad y(t) = Cq(t) + Du(t)$$

$$q(t) = \begin{bmatrix} i(t) \\ \omega(t) \end{bmatrix}, \qquad u(t) = \begin{bmatrix} v(t) \\ \tau_L(t) \end{bmatrix}$$

$$A = \begin{bmatrix} -4 & -0.2 \\ 5 & -5 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 0 & -50 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

1. Write the two scalar state equations. Explain the A entries, the two B columns and the load-torque sign. State the output and compare the model eigenvalues with the motor plant poles.

Optional extension Delay and phase margin

Unassessed optional practice. Add a pure delay of 0.050 s to the nominal loop.

1. Calculate the phase loss and remaining phase margin. Compare with the 45° target.

Submission and marking

Submit one individual report arranged by Stage and question. Include calculations and completed tables.

Include your new plots and modified scripts for **Stage 3, Questions 2–6**. Label axes with units; use clear legends and contrasting line styles.

Identify supplied material and any software or AI used.

Due: one week after core Class 8 at the published cutoff. Required questions: 100 marks. See the Optional bonus section for bonus marks.

Table 4. Stage marks (100 marks total).

Stage	Marks
Stage 1	20
Stage 2	30
Stage 3	50
Total required marks	**100**

Method and sources

Adapted from current MECH321 Lectures and Tutorials. Parameters and limits are teaching data; curves are model predictions.

Prof. Zengxi (Stephen) Pan | University of Wollongong | TYUT 2026 | October 2026